

Here are four key state management approaches in Flutter:

1. Stateless Widget (No State)

- Used for UI that does not change.

```
class MyStatelessWidget extends StatelessWidget {  
  @override  
  Widget build(BuildContext context) {  
    return Text("Hello, Flutter!");  
  }  
}
```

2. Stateful Widget (setState())

- Used for UI that needs to update dynamically.

```
class MyStatefulWidget extends StatefulWidget {  
  @override  
  _MyStatefulWidgetState createState() => _MyStatefulWidgetState();  
}  
  
class _MyStatefulWidgetState extends State<MyStatefulWidget> {  
  int counter = 0;  
  
  void increment() {  
    setState(() {  
      counter++;  
    });  
  }  
  
  @override  
  Widget build(BuildContext context) {  
    return Column(  
      children: [  
        Text('$counter'),  
        ElevatedButton(onPressed: increment, child: Text("Increment"))  
      ],  
    );  
  }  
}
```

3. Provider (Recommended for Simplicity)

- Efficient for managing global state.

```
class Counter with ChangeNotifier {  
  int _count = 0;  
  int get count => _count;  
  
  void increment() {  
    _count++;  
    notifyListeners();  
  }  
}
```

```
void main() {  
  runApp(  
    ChangeNotifierProvider(  
      create: (context) => Counter(),  
      child: MyApp(),  
    ),  
  );  
}
```

```
class MyApp extends StatelessWidget {  
  @override  
  Widget build(BuildContext context) {  
    final counter = Provider.of<Counter>(context);  
    return MaterialApp(  
      home: Scaffold(  
        body: Column(  
          children: [  
            Text('${counter.count}'),  
            ElevatedButton(  
              onPressed: counter.increment,  
              child: Text("Increment"),  
            ),  
          ],  
        ),  
      ),  
    );  
  }  
}
```

4. Bloc (Popular for Large Apps)

- Best for complex state management using events.

```
class CounterCubit extends Cubit<int> {  
  CounterCubit() : super(0);  
  void increment() => emit(state + 1);  
}
```

```
class CounterScreen extends StatelessWidget {  
  @override  
  Widget build(BuildContext context) {  
    return BlocProvider(  
      create: (context) => CounterCubit(),  
      child: BlocBuilder<CounterCubit, int>(  
        builder: (context, count) {  
          return Column(  
            children: [  
              Text('$count'),  
              ElevatedButton(  
                onPressed: () => context.read<CounterCubit>().increment(),  
                child: Text("Increment"),  
              ),  
            ],  
          );  
        },  
      ),  
    );  
  }  
}
```